

The sign of λ : an algebraic certificate for how a graph tower is generated

Ruqing Chen

GUT Geoservice Inc., Rivière-Beaudette, Québec J0P 1R0, Canada
ruqing@hotmail.com

21 September 2026

Abstract

For a tower of finite graphs $Y_0 \leftarrow Y_1 \leftarrow \dots$ the ℓ -adic size of the sandpile group often obeys a three-parameter law $\text{ord}_\ell |\text{Jac}(Y_k)| = \mu \ell^k + \lambda k + \nu$. In the established theory of abelian ℓ -towers this law is a theorem and μ, λ are non-negative by construction, being the Iwasawa invariants of a torsion $\mathbb{Z}_\ell[[T]]$ -module. We observe that the tower of iterated line digraphs — the standard design iteration for de Bruijn and Kautz interconnection networks — obeys the same law with $\lambda = -1$, on every base graph we tested, with $\mu = n(q+1)/q$ exactly. Since $\lambda < 0$ cannot be the λ -invariant of any torsion $\mathbb{Z}_\ell[[T]]$ -module, a measured $\lambda < 0$ certifies that the family is not an abelian ℓ -tower. We also find that the certificate detects *normality*, not commutativity: dihedral (non-abelian but Galois) towers give $\lambda = 3 > 0$, and only the non-normal line-digraph tower goes negative. We are careful about what is new. The abelian law is due to Vallières and collaborators; the line-digraph tower is classical in network design (Fiol–Alegre–Yebra, 1983) and its spanning-tree arithmetic is Knuth’s, in the sandpile form of Levine; indeed we show that $\lambda = -1$ telescopes directly from that classical recursion, and verify the telescoping on five bases. What appears to be new is organising the iteration as a tower, extracting (μ, λ, ν) from it, obtaining a negative λ at all, and reading the -1 as an Euler characteristic — the one-dimensional Eisenstein kernel removed at each level. Since the spanning-tree count is a standard network reliability measure, the law is a scaling law for a reliability measure whose linear coefficient records the design method. All 27 towers are computed in exact integer arithmetic.

1 Introduction

Given a family of networks that grows by refinement — a sequence $Y_0 \leftarrow Y_1 \leftarrow \dots$ in which each term covers or refines the last — one may ask how the family was generated. Existing answers in network science are statistical (likelihood, cross-validation, graphlet summaries) or geometric: the renormalisation exponents of Song, Havlin and Makse [19, 20] infer a growth mechanism from how box-covering statistics scale. This note offers an *algebraic* answer, in a narrow but exactly computable setting.

The invariant is the sandpile (critical) group $\text{Jac}(Y)$ [3], whose order is the number of spanning trees — itself a classical measure of network reliability [4, 7, 1]. Along a tower, $\text{ord}_\ell |\text{Jac}(Y_k)|$ frequently obeys

$$\text{ord}_\ell |\text{Jac}(Y_k)| = \mu \ell^k + \lambda k + \nu \quad (k \text{ large}). \quad (1)$$

What is established. For abelian ℓ -towers, (1) is a theorem. Vallieres [21] proved it for a family of towers over bouquets; McGown and Vallieres [18] proved it over an arbitrary connected multigraph; Gonet [13] obtained the Sylow structure from the $\Lambda = \mathbb{Z}_\ell[[T]]$ -module decomposition, and Kleine and Müller [15] and others have since extended the picture to $\mathbb{Z}_\ell^d$ -towers, ramified towers, weighted graphs and p -adic Lie voltage groups. In every one of these constructions the tower is a *voltage* (derived-graph) tower, hence Galois by construction, and $\mu, \lambda \geq 0$ because they are the Iwasawa invariants of a finitely generated torsion Λ -module [22]. The non-negativity is stated flatly throughout that literature.

Separately, the tower of iterated line digraphs is classical. Fiol, Alegre and Yebra [10, 11] introduced line-digraph iteration as *the* design method for dense interconnection networks, which is how de Bruijn and Kautz networks are built; and the spanning-tree arithmetic of one iteration is Knuth's [16], refined by Levine [17] to the group statement $\text{Jac}(G) \cong \text{Jac}(LG)/(q\text{-torsion})$, with the de Bruijn and Kautz cases worked out in [2, 5].

What is here. These two bodies of work have not met. We put the line-digraph iteration into the form (1) and find

$$\mu = \frac{n(q+1)}{q}, \quad \lambda = -1, \quad \nu = \text{const}, \quad (2)$$

on every base graph tested (Section 3). Since $\lambda < 0$ is impossible for a torsion Λ -module, this is a certificate (Section 4). We show in Section 5 that (2) telescopes from the classical recursion, so the arithmetic is not new — only its reading is — and we give the Euler-characteristic interpretation of the -1 . Section 6 draws the network consequence.

What we do not claim. Not (1) in the abelian case [21, 18, 13]; not the line-digraph tower as an object [10, 11]; not its spanning-tree counts [16, 17, 2, 5]. A referee will observe that (2) follows from [16, 17], and will be right; we say so in Section 5 and verify it.

2 Setting

A *tower* is a sequence of finite connected graphs with maps $Y_{k+1} \rightarrow Y_k$. We consider four generating mechanisms.

- (a) *Abelian*: Y_k is the derived graph of a voltage assignment in $\mathbb{Z}/\ell^k$, so the tower is Galois with group $\mathbb{Z}/\ell^k$.
- (b) *Rank two*: voltages in $(\mathbb{Z}/\ell)^k$.
- (c) *Non-abelian but normal*: voltages in the dihedral group D_{ℓ^k} .
- (d) *Non-normal*: $Y_{k+1} = L(Y_k)$, the line digraph, starting from the non-backtracking digraph of a base graph. Level- k vertices are the non-backtracking k -paths; this is the Fiol–Alegre–Yebra iteration [10].

For (a)–(c) $|\text{Jac}(Y_k)|$ is the number of spanning trees; for (d), which is a digraph tower, it is the number of spanning arborescences. Both are computed exactly as integer determinants.

3 Measurements

Table 1 reports (μ, λ, ν) fitted from the last three levels of each tower — fitting from the tail is deliberate, since the defects of (1) are finitely supported [6, IX] and the first levels may be transient — together with the number of levels on which the fit holds.

mechanism	instances	λ observed	law holds?
(a) abelian $\mathbb{Z}/2^k$	15 towers, 6 bases	1, 3, 5, 9 (all > 0)	yes
(b) rank two $(\mathbb{Z}/2)^k$	3 towers	—	no (as required)
(c) dihedral D_{2^k}	4 towers	3	yes
(d) line digraph	5 bases	−1 (every base)	yes

Table 1: The four mechanisms. For (b) the one-variable law (1) fails on all three towers, as it must: a $\mathbb{Z}_\ell^2$ -extension needs the two-variable theory of [8, 9, 15]. All entries exact.

base Y_0	n	q	μ measured	$n(q+1)/q$	$\text{ord}_q \text{Jac}(Y_k) , k = 1, 2, \dots$
K_4	4	2	6	6	7, 18, 41, 88, 183, 374
$K_{3,3}$	6	2	9	9	5, 22, 57, 128, 271
Q_3 (cube)	8	2	12	12	14, 37, 84, 179
Petersen	10	2	15	15	13, 42, 101, 220
$C_4 \times K_2$	8	2	12	12	14, 37, 84, 179

Table 2: The non-normal (line-digraph) towers. In every case $\lambda = -1$ and μ equals $n(q+1)/q$ exactly, and the fitted law reproduces every computed level. All entries exact.

The contrast in Table 1 is the point, and it is not the contrast we expected. Negative λ does not track non-commutativity: the dihedral towers are non-abelian and give $\lambda = 3 > 0$. It tracks *normality*. Every tower in the graph-Iwasawa literature is a voltage tower and hence Galois; the line-digraph tower is not, and it is the only one that goes negative.

4 The certificate

Theorem 1. *Suppose a tower satisfies (1) with $\lambda < 0$. Then the tower is not an abelian ℓ -tower: there is no voltage assignment in $\mathbb{Z}_\ell$ whose derived graphs are the Y_k .*

Proof. For an abelian ℓ -tower the inverse limit $X_\infty = \varprojlim_k \text{Jac}(Y_k) \otimes \mathbb{Z}_\ell$ is a finitely generated torsion $\Lambda = \mathbb{Z}_\ell[[T]]$ -module [6, IV], [13, 18], and its Iwasawa invariants satisfy $\mu = \sum_i k_i \geq 0$ and $\lambda = \sum_j m_j \deg g_j \geq 0$ by the structure theorem [22, Thm. 13.12]; these are the coefficients in (1). $\square$

Remark 2 (What the certificate does not say). Theorem 1 rules out the characteristic ideal of a *single* torsion Λ -module. It does not rule out a virtual object — a formal difference of two Λ -modules, or the cohomology of a complex — and indeed the interpretation in Section 5 is exactly of that kind. It is therefore wrong to say that $\lambda = -1$ contradicts Iwasawa theory; the correct statement is that it is the invariant of a difference, not of a module. Two further limitations: the certificate is one-directional (Table 1(c) shows non-abelian towers with $\lambda > 0$), and it presupposes that (1) holds at all, which for rank-two towers it does not.

Remark 3 (A parity coincidence). Kataoka [14] shows that every pair (λ, μ) with λ odd is realised by an unramified $\mathbb{Z}_\ell$ -covering of a bouquet. Our $\lambda = -1$ is odd. We have no reason to think this is more than a coincidence, but it is the kind of thing a reader notices, so we record it.

5 Where the -1 comes from

Proposition 4. *Let Y_1 be a q -regular Eulerian digraph on n_1 vertices and $Y_{k+1} = L(Y_k)$. The classical line-digraph tree formula [16], in the form used in [6, III] and equivalent to Levine's $\text{Jac}(G) \cong \text{Jac}(LG)/(q\text{-torsion})$ [17], gives*

$$\text{ord}_q |\text{Jac}(Y_{k+1})| - \text{ord}_q |\text{Jac}(Y_k)| = n_k(q-1) - 1.$$

With $n_k = n(q+1)q^{k-1}$ this telescopes to (2).

Proof. Summing the increment from level 1 to level k gives $n(q+1)(q^{k-1} - 1) - (k-1) + \text{ord}_q |\text{Jac}(Y_1)|$, which is $\frac{n(q+1)}{q}q^k - k + \text{const.}$ $\square$

So $\lambda = -1$ is a consequence of a 1967 formula. We verify the telescoping numerically on all five bases (check (K) of Section 7): the measured sequences of Table 2 agree term by term with the telescoped prediction.

The Euler characteristic. The -1 in the increment is not a normalisation artefact. The one-step increment is $n_k(q-1) - \chi_k$ with $\chi_k = \dim \ker \Delta_k^{\text{dir}} = 1$: at every level a one-dimensional kernel — the constant, or Eisenstein, direction — is removed [6, III]. Summing a constant Euler characteristic over k levels is what produces a term linear in k with negative coefficient, and it is why no single Λ -module can have this as its characteristic ideal: the object is a difference, one term of which is the Eisenstein line. Check (E) verifies the increment identity level by level.

6 The network reading

The two mechanisms of Table 1 are not artificial. Torus, circulant and chordal-ring interconnects are voltage covers of a small base, hence abelian towers; de Bruijn and Kautz interconnects are built by line-digraph iteration [10, 11, 12], hence non-normal towers. The quantity being measured, $|\text{Jac}(Y_k)|$, is the spanning-tree count, a standard reliability measure [4, 7, 1]. Equation (1) is therefore a three-parameter scaling law for a reliability measure along a design family, in which

- μ is the bulk growth rate — for the line-digraph family, $n(q+1)/q$, i.e. proportional to the level-one size;
- λ records the design method: positive for covering constructions, -1 for line-digraph iteration;
- ν is an offset with no evident structural meaning.

We are not aware of prior use of an algebraic growth exponent for this purpose; the closest precedent is the use of renormalisation exponents to infer a growth mechanism [20]. Whether the law survives the perturbations a real network family suffers — irregularity, missing edges, coarse-grainings that are not covers — we have not tested, and Section 8 lists it first among the open questions.

7 Verification

The script `tower_lambda.py` accompanying this note builds and measures 27 towers and runs the checks of Table 3 in about 7 minutes, all passing; its output is archived as `tower_lambda_output.txt`. All spanning-tree and arborescence counts are exact integer determinants.

key	check	mode
(A)	(1) holds with $\lambda > 0$ on 15 abelian towers over 6 bases	exact
(L)	$\lambda = -1$ and $\mu = n(q+1)/q$ on 5 line-digraph towers (Table 2)	exact
(E)	the increment is $n_k(q-1) - \chi$ with $\chi = 1$, level by level	exact
(K)	Prop. 4: the measured sequence equals the telescoped recursion	exact
(R)	the one-variable law fails on all 3 rank-two $(\mathbb{Z}/2)^k$ towers	exact
(D)	dihedral towers: $\lambda = 3 > 0$, so the certificate detects normality	exact
(C)	the certificate assembled over the whole battery	exact

Table 3: The verification battery (`tower_lambda.py`), 7/7 passing.

8 Scope

Quoted. The abelian law [21, 18, 13, 15]; non-negativity of Iwasawa invariants [22]; the line-digraph iteration [10, 11]; its tree arithmetic [16, 17, 2, 5]; spanning trees as a reliability measure [4, 7, 1].

New here, as far as we can determine. Putting the line-digraph iteration in the form (1); the value $\lambda = -1$, there being no negative λ anywhere in the graph-Iwasawa literature; the Euler-characteristic reading of it; the certificate (Theorem 1) with its caveats (Remark 2); and the observation that the certificate separates normal from non-normal rather than abelian from non-abelian.

Open. (a) Robustness. Every tower here is exact; real network families are not, and we do not know how (1) degrades under edge noise or under refinements that are not covers. This is the question that decides whether any of this is usable. (b) Whether $\lambda < 0$ occurs for any tower other than the line-digraph family — we found none, but tested only four mechanisms. (c) Whether the structural refinement of λ as a count of persistently growing cyclic factors is already contained in [13] or in Kataoka’s work on Fitting ideals; we were not able to settle this. (d) The rank-two non-normal case, where [6, X–XII] shows no abelian sector exists at all.

Provenance

Unrefereed preprint. Theorem 1 is a restatement of the structure theorem for Λ -modules and is proved above modulo that theorem. Proposition 4 is a summation of a classical formula. Everything numerical is produced by the accompanying script in exact integer arithmetic. We searched for a negative λ , for a non-normal graph tower with a growth law, and for an Euler-characteristic reading of such a law, and found none; a search cannot establish a negative, and our coverage of engineering proceedings in particular is incomplete.

References

- [1] J. S. Baras and P. Hovareshti, *Efficient and robust communication topologies for distributed decision making in networked systems*, Proc. 48th IEEE Conf. Decision and Control (2009).
- [2] H. Bidkhori and S. Kishore, *Counting the spanning trees of a directed line graph*, Combin. Probab. Comput. **20** (2011), 11–25.
- [3] N. Biggs, *Chip-firing and the critical group of a graph*, J. Algebraic Combin. **9** (1999), 25–45.
- [4] F. T. Boesch, *On unreliability polynomials and graph connectivity in reliable network synthesis*, J. Graph Theory **10** (1986), 339–352.
- [5] S. H. Chan, H. D. L. Hollmann and D. V. Pasechnik, *Sandpile groups of generalized de Bruijn and Kautz graphs and circulant matrices over finite fields*, J. Algebra **421** (2015), 268–295.
- [6] R. Chen, *Algorithmic non-commutative class field theory, I–XVI*, preprints (2026); here III (the line-digraph recursion and the Euler term), IV (the limit module), IX (the defects), X–XII (rank two).
- [7] C. J. Colbourn, *The Combinatorics of Network Reliability*, Oxford University Press, 1987.
- [8] A. A. Cuoco and P. Monsky, *Class numbers in $\mathbb{Z}_p^d$ -extensions*, Math. Ann. **255** (1981), 235–258.
- [9] S. DuBose and D. Vallières, *On $\mathbb{Z}_\ell^d$ -towers of graphs*, preprint, arXiv:2207.01711 (2022).
- [10] M. A. Fiol, I. Alegre and J. L. A. Yebra, *Line digraph iterations and the (d, k) problem for directed graphs*, Proc. 10th Int. Symp. Computer Architecture (1983), 174–177.
- [11] M. A. Fiol, J. L. A. Yebra and I. Alegre, *Line digraph iterations and the (d, k) digraph problem*, IEEE Trans. Comput. **C-33** (1984), 400–403.
- [12] M. A. Fiol and A. S. Lladó, *The partial line digraph technique in the design of large interconnection networks*, IEEE Trans. Comput. **41** (1992), 848–857.
- [13] S. R. Gonet, *Iwasawa theory of Jacobians of graphs*, Algebraic Combinatorics **5** (2022), 827–848.
- [14] T. Kataoka, *Construction of graph coverings with prescribed Iwasawa invariants*, preprint, arXiv:2603.23088 (2026).
- [15] S. Kleine and K. Müller, *On the growth of the Jacobian in $\mathbb{Z}_p^l$ -voltage covers of graphs*, Algebraic Combinatorics **7** (2024), no. 4.
- [16] D. E. Knuth, *Oriented subtrees of an arc digraph*, J. Combin. Theory **3** (1967), 309–314.
- [17] L. Levine, *Sandpile groups and spanning trees of directed line graphs*, J. Combin. Theory Ser. A **118** (2011), 350–364.
- [18] K. McGown and D. Vallières, *On abelian ℓ -towers of multigraphs III*, Ann. Math. Québec **48** (2024), 1–19.

- [19] C. Song, S. Havlin and H. A. Makse, *Self-similarity of complex networks*, Nature **433** (2005), 392–395.
- [20] C. Song, S. Havlin and H. A. Makse, *Origins of fractality in the growth of complex networks*, Nature Physics **2** (2006), 275–281.
- [21] D. Vallieres, *On abelian ℓ -towers of multigraphs*, Ann. Math. Québec **45** (2021), 433–452.
- [22] L. C. Washington, *Introduction to Cyclotomic Fields*, 2nd ed., Graduate Texts in Mathematics **83**, Springer, 1997.